

CHAPTER 5

BRAKING AND ENTERING

IN THE CHAPTER ON THE three basics, we introduced the idea that you could break down the corner entry process into four main blocks, and that you could use some or all of these

blocks, depending on the corner entry situation.

- Block 1: the throttle-brake transition. To use the brakes you have to take your foot off the throttle and put it on the brake pedal.
- Block 2: straight-line deceleration.
- Block 3: brake-turning.
- Block 4: the brake-throttle transition, where you switch from the brake pedal back to the throttle.

In terms of reducing lap time, this last of the three cornerstones of a fast lap has less to contribute since it represents a much smaller segment of the course than the other two. Nonetheless, the braking and entering section of the race lap is typically where the best drivers find the last second of lap time.

You know that one technique does *not* fit all, but how do you choose between the alternatives? By using the building-block approach to closely analyze skills, you can determine the type of corner entry technique you should use for a particular setting. This is especially helpful when you're faced with a new racetrack and have to decide on how to approach each corner entry. Let's start with Block 1.

BLOCK 1: THROTTLE-BRAKE TRANSITION

Coming off of the throttle and onto the brakes at the end of a long straightaway is a vastly different process than you're accustomed to using in your everyday driving. On the street, you're trying to be smooth with the car and sensitive to your passengers, so you don't move your foot from the throttle to the brake with much speed or force.

In a racecar approaching a 35 m.p.h. corner at 110 m.p.h., as you would be in one of our Formula Dodge

There's more to corner entry than the simplistic "brake late, brake hard" approach. Four "building blocks" make up the corner entry process. The speed loss required for the corner determines when and how you use these blocks.

cars at the Sebring hairpin, you don't want to waste any time getting off the throttle and onto the brake pedal. You'll carry the straightaway speed at full throttle right up to your chosen braking point

and instantly get off the throttle and onto the brakes—hard. There's nothing gentle about it.

No Slams

You go to a lot of braking force in a very short period of time in racing. In our Formula cars at Sebring, a good driver will go from full throttle to over 130 lbs. of foot pressure on the brake pedal in 3/10 second. You would think that you could only describe this as "slamming on" the brakes, but that's not the case. Slam implies impact. Here, you're moving your foot from the throttle to the brake pedal *fast*, but the build up of pressure is a hard squeeze as opposed to a "slam."

Slamming on the brakes is a common mistake. Some drivers hit the brake pedal as if driving a nail with a hammer. This often results in instantaneous front wheel lockup. This lockup occurs because the front brakes start slowing down the front tires before load transfer can help deliver extra grip.

Fleet Feet

In straight-line threshold braking, there is going to be little variation in the move from throttle to brake. You want to spend as little time as possible getting off the throttle and into significant brake pressure to avoid wasting time coasting. When the speed loss requires less than threshold, it still makes sense to go to the brakes as quickly as possible.

There are exceptions. If the car you're driving is very sensitive to pitch changes—for example has idiosyncrasies that exaggerate the loss of rear wheel traction in a throttle lift—you might breathe the throttle back more gently before going to the brake pedal.

"In braking a Formula car, you have almost instantaneous feedback through the tires and suspension—also visually. In a showroom stock car it's a little softer—everything takes more time. It's like night and day, really. In a Formula car you just get with it. In the Saturn I raced, you settle it, then go at it."

—Robbie Buhl

Throttle-Brake While Turning

In some corners, the first application of brakes comes *after* you've turned into the corner. In these cases you have to be careful about snapping off the throttle on the way to the brakes. If the car is cornering hard, the sudden loss of rear tire traction caused by an abrupt lift off the throttle can cause a spin. As you'll see later, the throttle-brake transition under these circumstances has to be done more gradually, and with a good degree of sensitivity and anticipation.

BLOCK 2: STRAIGHT-LINE DECELERATION

Going from the throttle to the brake is just the start of the straight-line deceleration braking block. As you've seen, you don't want to *slam* the brake pedal but you do want to squeeze the brakes on hard, quickly.

Not Too Delicate

We see our share of gorilla brakers, but a far more common mistake for beginning drivers is to underestimate just how hard you can push on the brake pedal without locking up the tires. A good street sport sedan like a Dodge Stratus requires only 40 lbs. of pressure on the brake pedal to generate 80% of its maximum braking force. This level of braking would be considered quite aggressive for normal street driving. A driver transitioning into racecars understandably has to relearn the pedal pressure effort required near the limit.

Whoops!

To locate the limit, you're going to have to push harder and harder on the brake pedal until at some point there is too much pressure and the tire locks.

In our racing school we devote a considerable amount of time, and a lot of Michelin's rubber, to allowing fledgling racers to experiment with the limit under threshold braking. You can intellectualize all you like, but there is no substitute for putting the brakes on hard 100 times to learn something about what's required.

In these exercises the side benefit is plenty of practice in unlocking a locked tire. You know that a locked tire has 30% less traction than one at its brak-

ing limit, so you need to relax enough pedal effort to get the tire turning again. The difficult talent to develop is the feel for just how much pressure to relax.

Learning to Modulate

Brake modulation involves both increases and decreases in brake pedal pressure. You should be adjusting pedal pressure to compensate for changes in traction between the tire and road surface, constantly flirting with lockup—adding pressure to probe the limit, relaxing pressure at the hint of lockup. That's when it's done perfectly, but there are lots of ways to fall short of perfection.

The gross mistakes fall into two categories. The first is when you overreact to a locked tire and leap off the brake pedal. The car—which, because of the locked tire, has been slowing at 70% of its potential—now loses *all* of its braking force. Think of the logic here—giving away *all* your braking in an attempt to recover 30%. Normally this total release of pressure is followed by another application of brakes and another lockup, soon followed by another total release.

This pumping of the pedal is playing havoc with the car's load distribution. First the nose is down, then it leaps up as the brakes are released, then down again as the brakes go back on.

When it comes to brake modulation changes in *pressure* rather than changes in *dimension* are required. For example, to unlock a locked tire in the latest version of our Formula Dodge, the brake pedal pressure needs to be released from approximately 140 lbs. down to 100 lbs. The foot motion required to effect this 40 lb. change in pressure on the face of the pedal is barely perceptible to the naked eye. So, instead of changing the brake pedal pressure by moving your entire leg, try changing the level of pressure by changing the tension of the muscles in your lower leg and ankle. The changes are more likely to be subtle, which is exactly the type of change needed.

The second gross error is the tendency to respond to wheel lockup by continuing to increase brake pedal pressure on up to the point where you could create

"No one, absolutely no one, is proof against panic. More than 20 years ago, the late Graham Hill, twice world champion racing driver and one of the most expert and experienced drivers alive, stuffed one of our Ford GT40s into an earthen bank at Le Mans, doing considerable damage. When I asked him what happened, he replied, 'I went into the corner a little too fast, realized it, called upon my legendary skill and years of experience, panicked, locked up the brakes and slid into the bloody bank—sorry!'"

—Carroll Smith

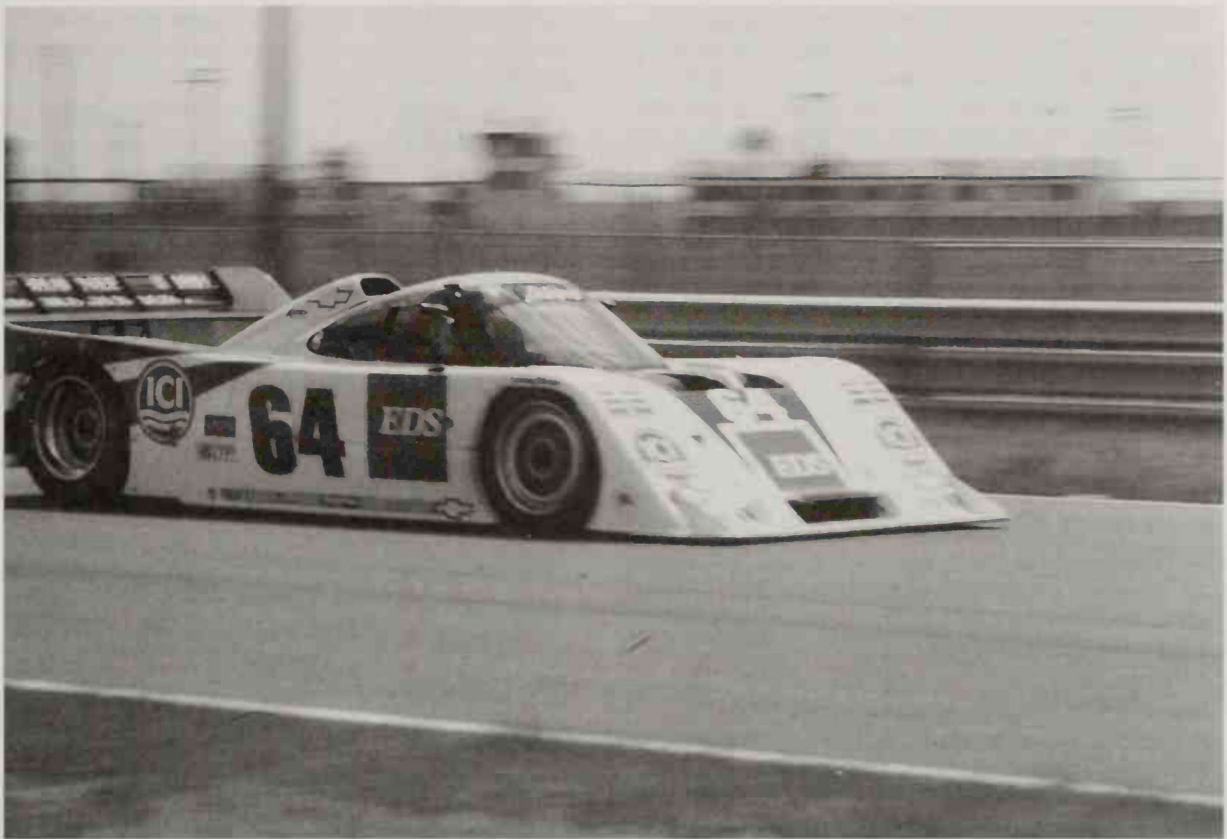


Fig. 5-1. The deceleration rate of a race car with ground effects, like the GTP Intrepid pictured here, can approach over 80 m.p.h. per second.

diamonds from coal. This seems to happen with greater frequency when there is a barrier or other solid object in your line of sight at the end of the braking zone. There's an element of panic in this mistake. It is also the kind of error at which experienced drivers might sneer. But beware smugness—no one is ever immune to these kinds of errors, no matter how experienced they are.

Gaining Time In Long Braking Zones

When you are driving a racecar as fast as it will go in a long straight-line braking zone, it's at the absolute limit of braking, at the threshold of lockup. As we mentioned earlier, this is only half of the aim of minimizing time in this section of the racetrack. The other, equally important part is arriving at the turn-in point of the corner at just the right speed—neither too fast for the car to be able to turn into the corner on the right line, nor so slow that you're not using all of the grip available.

Easy to say, but it's not reasonable to expect a relatively inexperienced driver to reach this goal right out of the box. Even a skilled racer doesn't get it perfectly right on the first few attempts.

You need to come up with ways to approach the limit in heavy braking in a way that can whittle the

time down without taking too big a bite, risking an excursion off the racetrack.

"I have to use specific brake points. You just can't be consistent enough using any other method. At least I know I can't."

—Jeremy Dale

Brake Points

You spent a good deal of time finding reference points to help keep you on the proper path through the corners. In braking you need to develop reference points to cue your application of the brakes. Racetracks are generally cooperative about putting number boards up in the obvious hard braking zones, but, more often than you would like to think, there are no clear markers to help you apply the brakes at the same spot every lap. Nonetheless, you have to seek out a brake point at the entry of every corner that will require braking.

The main difficulty is that brake points are variable. If you're taking it easy, learning the line around the racetrack, your straightaway speeds are going to

be lower. Consequently, you can put the brakes on *later* than you will when you start flying. On the other hand, you'll tend to be on the brakes *earlier* because most of your concentration is on getting the line and corner exits right before trying to get the last bit of lap time from braking skills. The key is to establish a brake point for every corner and then work off of that point to suit the circumstance.

Don't Start By Going Deeper

The most common mistake that drivers make when they turn their attention to getting the last bit of lap time available at corner entries is to drive closer to the corner before braking—going deeper.

Going deeper, first of all, should be reserved for high-speed-loss corners where threshold braking will be used. It is very seldom that threshold braking will be the appropriate technique to use in every corner entry situation on the track. At Lime Rock, for example, there are six corner entries for every lap; only one involves threshold braking, and there's some active discussion about that one.

Identify the Threshold—"The Procedure"

In a threshold braking zone, it doesn't make sense to start going deeper before you are certain that you're using maximum braking. The first step in a procedure for making up time on corner approaches is to push harder and harder on the brake pedal to identify where the threshold is. Once you get to the point where you lock up a tire occasionally, you can start to move the brake point closer to the corner.

By starting out the process using more braking force, you actually lower the speed at the turn-in since you're braking over the same distance but with more force. Once you've identified the right level of force, you raise the speed at the turn-in point by moving the brake point closer to the corner—a little at a time. At some point you will have gone too far; your entry speed will be such that the car won't turn on an arc that will get it to the Apex. The car will still turn: it just won't be able to run on the best line. You'll probably need to continue the brake-turning portion of the corner entry to the point where it interferes with the throttle application, hurting your corner exit and the following straightaway.

"I don't brake at the absolute possible last moment every corner of every lap because to do so, you're going to give up a bit at the exit by virtue of the fact that you can't pick up the throttle where you might otherwise be able to."

"If you're braking at the last possible moment, that means braking to, or slightly past the apex. To do that, you sacrifice the exit."

—Bryan Herta

"The Procedure" for working your way up to a quicker corner entry of a threshold braking corner is straightforward:

- 1) Identify the level of threshold braking available by braking harder and harder on successive laps.
- 2) Once you're in the vicinity of threshold braking, move the brake point down toward the corner in small increments (three feet at a time is a good rule of thumb).
- 3) If entry speed gets so high that the car begins to be forced off-line or the throttle-application is delayed, you've taken the braking *too deep*. Move the brake point back to where it allowed you to do the line and corner exit properly.

"I absolutely use 'The Procedure' we teach—all the time. I can't see that there's any other way to approach it. If you try to brake late before you've ID'd the brake level available, you're wasting time. You start by braking a little early and working on the level that's available to you. Once you've ID'd how much is there you can start working it down. As you get more experience and confidence you can do both—you can step the brake level up and move it down toward the corner at the same time."

—Jeremy Dale

Always Start With Threshold?

In a long braking zone, threshold braking from top speed down to the maximum speed the car will take at the turn-in point is the fastest way. No argument. But this isn't necessarily true in all braking zones.

The deceleration from threshold braking is formidable. In one of our Formula cars, the deceleration is over 1 G: at least 22 m.p.h. per second. Some racecars with slicks, wings and downforce-producing underbodies can see deceleration as high as 4 Gs, or 88 m.p.h. per second! In a car such as this, a corner entry situation requiring an 8 m.p.h. loss of speed would take under .1 seconds of threshold braking. Many drivers are tempted to use threshold braking under all situations, but this outlook is a little too simplistic.

Words of Advice

When Mario Andretti was competing on the Formula One circuit for Lotus, the team for which he won his World Championship, he was asked by an interviewer to talk about his driving technique. His response was that his technique was his livelihood and something that he might discuss with his sons but not a subject for public comment. "I will say one thing," he volunteered, "It is amazing how many drivers, even at the

Formula One level, think that the brakes are for slowing the car down."

The enigmatic quality of that statement is priceless, and there's an underlying message that the braking process does more than just slow you down. It also has a significant effect on the attitude and balance of the car as it works its way to the throttle application point.

The Effect of Short, Sharp Braking

In a car capable of 4 Gs of braking force (an Indy Car), think of what a threshold brake application does to the loading and attitude of the car in this situation—full throttle to maximum braking to brake release then turn-in—all in a tenth of a second. You might as well hit the car with a sledgehammer. The better move would be to not use everything the car had available in braking and instead concentrate on losing just the right amount of speed. Using lighter braking over a longer period of time allows the car to be more predictable and stable as you turn into the corner.

You're shooting for a specific target entry speed. By using the brakes forcefully for such a short period of time it's hard to get the speed just right. The tendency would be to over-slow the car, losing speed that may never be recovered on the following straight.

One rule of thumb is that you should consider abandoning threshold braking when the speed loss at the corner entry requires less than a full second of braking. If the corner is one which doesn't require a downshift, think of using a level of straight-line braking less than threshold in order to balance the car at the corner entry at the precise maximum entry speed.

What Does Lighter Braking Cost?

For those die-hard threshold brakers out there, let's do some lap time comparisons of threshold vs. 75% braking.

In Fig. 5-2, you can see that at 75% of threshold it takes 66 feet to slow the car from 80 m.p.h. to 70 m.p.h. by the turn-in (you will do additional slowing *after* the turn-in point and before the throttle application point, but your target entry speed is 70 m.p.h.). It takes the car .6 seconds to cover this distance.

If you go to threshold braking, it will only take you 50 feet to lose the 10 m.p.h. and you'll cover the last 66 feet up to the turn-in in .572 seconds. That's just over a quarter of a tenth of a second difference.

By braking more lightly, you're gambling that the slight time sacrifice on the way into the corner will be made up by doing the 75% braking more consistently, giving you better control of the entry speed and car. With the car more in control you'll be more consistent with your throttle application point.

Comparing Different Strategies in Long Braking Zones

You've seen that using "The Procedure" is an established way to work on improving at high-speed-loss corner entries. In this approach you increase speed at the turn-in by braking later and later. Another way to accomplish the same goal is to keep the brake point consistent and alter the level of braking force. Let's take a look at some real life runs through the last 400 feet approaching the turn-in point of the Sebring hairpin to see how the time in the straight line braking zone varies using different approaches (see Fig. 5-3).

On the quickest run, the driver approaches the braking zone at 104 m.p.h. The driver lifts off the throttle, and it takes .15 seconds before the first bit of brake pressure. A tenth of a second later the pressure on the brake pedal is at 65 lbs., peaking at 122 lbs. in the next tenth. Threshold braking begins 339 feet before the turn-in point. It has taken .35 seconds to go from full throttle to maximum braking. During the de-

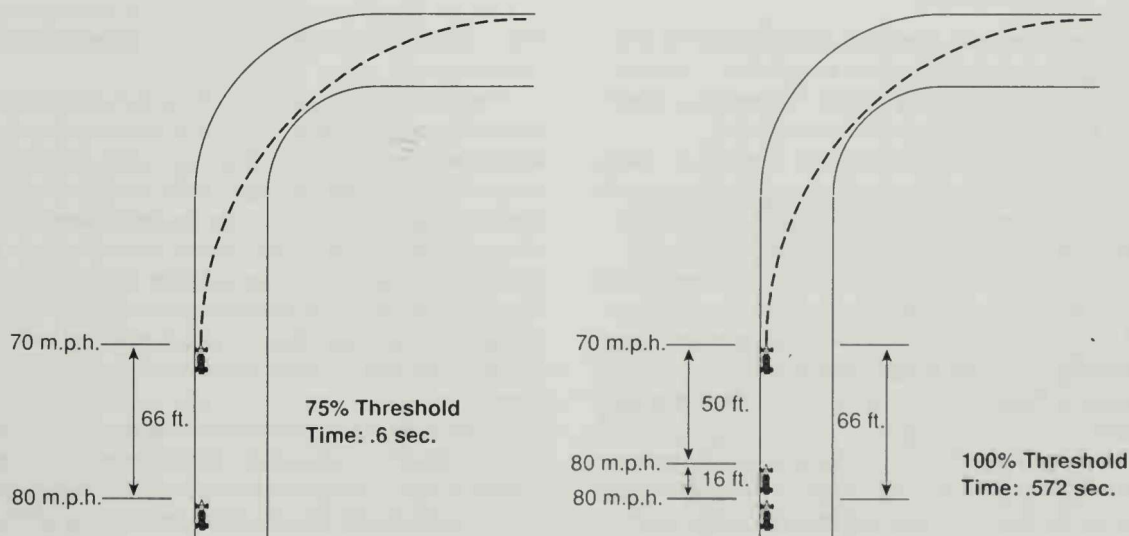


Fig. 5-2. In instances where the speed loss required at the corner entry is relatively small, braking short of threshold has a minor effect on potential lap time.

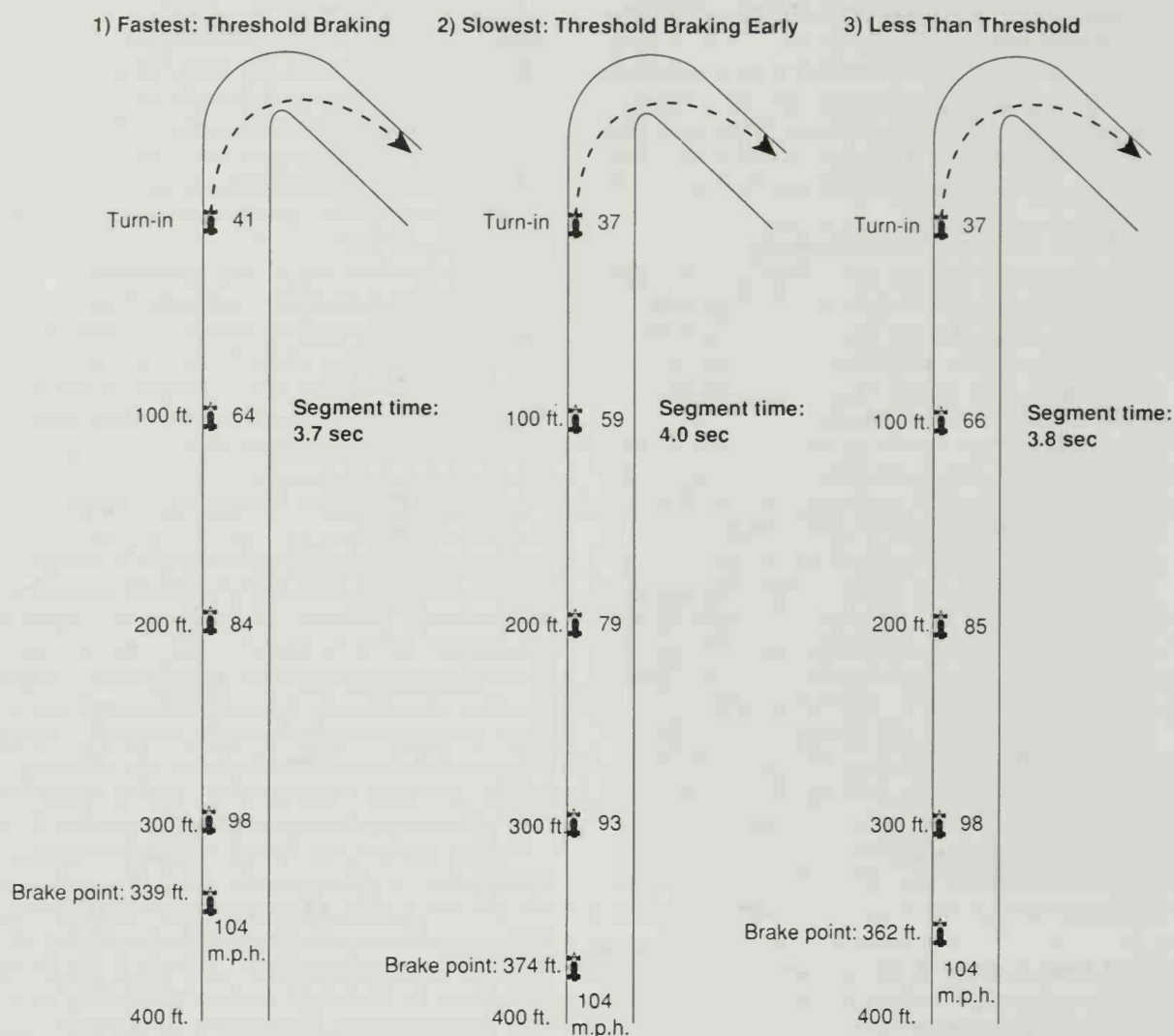


Fig. 5-3. Three different approaches to the last 400 feet of the Sebring straight, leading to the hairpin.

celeration, the brake pressure varies from 90 to 130 lbs. The driver gets to the turn-in point at 41 m.p.h., 3.7 seconds after braking begins. The average speed loss is 19.7 m.p.h. per second—nearly .9 G.

On the slowest run, which takes .3 seconds longer, braking started 374 feet from the corner, 35 feet earlier than in the fast run—in other words, about three car lengths earlier. The time it took to go from full throttle to maximum brake pressure was identical to the first run, as was the maximum brake pedal pressure. The rate of deceleration was likewise very similar, around .9 G. The biggest differences were where the braking began and the speed of the car at the turn-in point.

The surprising thing is that the .3 seconds were not lost by braking 35 feet early; in fact, the lap time differential in the first 35 feet is imperceptible. The difference in lap time comes from the run being from 3 to 4 m.p.h. slower over the last 339 feet of the braking

zone, culminating in being 4 m.p.h. slower at the turn-in point.

The problem here isn't a matter of not slowing the car fast enough—the deceleration rates are nearly identical—but that the driver started the process too soon. When the driver stayed with this hard level of braking all the way to the turn-in point, the car had the same speed shortfall at the corner entry. Actual data shows that, during the quicker run, the driver reduced the level of braking pressure in the last 100 feet of the braking zone. One conclusion is that the slower run is the ideal type to gain from following "The Procedure."

There is another way to work the time down. Let's look at the third run and see how much of the three tenths of a second deficit our driver can make up without driving 35 feet closer to the corner before going to the brakes.

In this pass, the braking starts 362 feet from the turn-in, and isn't as aggressive, running 15 to 30% lighter than the fast run. But the lighter braking keeps the car's speed, foot by foot, equivalent to the faster pass for about 3/4 of the braking zone. The time difference between these two runs is down to one tenth of a second.

That last tenth comes from the last 100 feet before the turn-in where the driver's slow build-up of braking pressure finally surpasses that of the faster run and the speed begins to fall below the target, ending up 4 m.p.h. slower at the turn-in.

What does all this mean? Hopefully, it can give you tools you can use to improve in hard braking zones. The fastest way to handle threshold braking situations is, just like the theory says, to find the absolute maximum level of braking force available and use it to slow the car from straightaway speed to the maximum possible turn-in speed for the corner entry.

In our examples, 41 m.p.h. is the fastest speed we have seen at the turn-in point with a skilled driver over a multitude of laps, so we can safely call this the limit—the fastest the car can be going at this point and still make the apex.

If your driving is less than perfect and you are losing time in the braking zone, you can see in this illustration that 2/3 of the time deficit can be made up by actually being below the braking threshold and braking earlier than required. The lighter level of braking keeps the car at a speed, foot by foot, on the approach to the corner, close to what the car would be doing with a later, harder brake application.

It's important to realize that this driver, in one of the hardest braking zones to be found anywhere, was just a tenth of a second off the peak performance while being 15 to 30% under the threshold. This reinforces the tactic of concentrating on line and exit speed first, and leaving the effort at braking at the absolute limit until later when that tenth of a second is an improvement you need.

Modulation—Not Just for Lockup

There are potential advantages to choosing not to brake at one consistently hard level from the time you touch the brakes all the way to the turn-in point. In the fastest run in the above example, the driver modulates away from threshold braking to a lower level about 90 feet short of the turn-in. It's an effective way of boosting the speed at the turn-in, while experimenting with the maximum speed available at the corner entry. This allows you to get acclimated to the sensation of speed at the turn-in point so that when it comes time to get every bit out of the corner entry, you have an appreciation for the speed potential at the turn-in point without the distraction of the effort it takes to be right on the threshold braking limit.

"A lot of drivers forget that you are not locked into one level of pressure on the brake pedal. Even though you feel that you may be a little early on the brakes, there's nothing wrong with that because you can always pull back five per cent on the pedal and kind of float the entry speed into the corner. The pressure can always be modulated."

—Jeremy Dale

Lighter Braking for Smaller Speed Losses

The last ten miles per hour of speed difference between approach speed and corner entry speed is best handled with progressively less and less braking force over approximately the same time and distance. You've seen how small a difference in lap time there is in situations where the speed difference is 10 m.p.h. The time spread is even smaller when you're in the 4 m.p.h. to 1 m.p.h. range. Look at Fig. 5-4 to sketch out how you might deal with the last 10 m.p.h. of straight-line speed loss.

Speed loss	Control input
10 m.p.h.	75% braking
9	60% braking
8	50% braking
7	30% braking
6	20% braking
5	Full-throttle lift
4	Shorter full-throttle lift
3	Partial throttle lift
2	Shorter breathe
1	The kind of lift only you and God know about
0	No breathe—full throttle

Fig. 5-4. Small speed losses at the corner entry can be accomplished with a variety of methods from light braking to a tiny lift off the throttle.

Using the Breathe

Just lifting off the throttle will slow the car down more gently than the brakes, and is often used to make small speed adjustments at the corner entry. Coming part way back from full throttle reduces the car's speed by removing some of the thrust that accelerates the car against the resistance of inertia and mechanical and aerodynamic drag. Lifting further creates de-

celeration as the motor creates a drag of its own on the rear tires.

There is a difference in these two types of throttle-related deceleration that is worth noting. The light breathe—one that reduces thrust—reduces the car's speed without making any demands on the tires. Lifting enough so that engine compression aids the slowing demands a portion of the traction available to the rear tires (in a rear wheel drive racecar). Normally, this is not a problem when the car is in a straight-line braking zone. As you've seen earlier, however, the car can be biased toward oversteer if the throttle is lifted while the car has cornering loads on it.

As the speed loss requirements go down, the full breathe will get shorter and shorter until, rather than a short, sharp lift, a partial breathe over a longer period of time will accomplish the same speed loss without detrimental effects on the balance of the car. And, as breathes get lighter and lighter, they enter the realm where they're doing more for your psyche than they are to solve an entry speed differential.

"One thing that has always amazed me is that the difference between running a fast corner flat out and running it with just the tiniest lift, is remarkable. Just a little breathe on the throttle puts enough weight on the front of the car so that it turns in really well, whereas without the lift it just doesn't want to turn."

—Jeremy Dale

BLOCK 3: DECELERATING WHILE TURNING (TRAIL-BRAKING)

As stated earlier, there has been, and still is, a philosophy that *all* slowing for corners should be done on the straight. We disagree. Since the Skip Barber Racing School began in 1975, we have trained drivers to use not only straight-line braking but also the brake-turning ability inherent in every vehicle. Our philosophy is that it is easier in the long run to learn the skill and sensitivity required to brake and turn right from the beginning rather than learn a technique that will then have to be modified at a future date. If you choose to continue braking past the turn-in point, be it 150 feet or two feet, we call this "trail-braking."

In reality, all the argument over whether to trail-brake or not is a bit over-blown. Using the braking and turning ability of a car is just another tool in the effort to lower lap time. You don't use a hammer for *every* repair, but it's the perfect tool for some jobs.

No one is sure who coined the phrase, but the first well-known driver to talk publicly about braking and turning was Mark Donohue. In the late 1960s, Mark

"This whole issue of trail-braking or not trail-braking is bullshit because every quick driver trail-brakes, whether they talk about it or not. On fast racetracks, like you find in Europe, you might do it less, but in the States you have more tight turns where you have to carry the brakes in there to help point the car and to gain an advantage by going deeper. There should be no question about it. That's that."

—Robbie Buhl

and other members of the Road Racing Drivers Club held seminars on racing technique and car preparation; it was in these seminars that Mark first began the heretical public discussion of using the brakes beyond the straight-and-narrow confines of conventional racing technique. The prevailing wisdom was that one was courting disaster by mixing the steering wheel and the brake pedal. Mark, a Brown University-trained mechanical engineer, based his premise on a graphic conceptualization of a tire's traction capabilities, which he called the "friction circle."

The Friction Circle

Let's build a friction circle graph. For the sake of simplicity we'll graph a 1 G tire. Under acceleration, this tire could create 1 G of accelerating force. Under braking the force would be the same, 1 G, but in the opposite direction. Cornered at its maximum, the tire would create 1 G of cornering force.

In Fig. 5-5, we graph acceleration on the vertical axis below the point where the two axes meet. The tire's braking ability is shown on the vertical axis above the 0 point. To the right of the origin is the tire's right cornering force; to the left, left cornering force. Since this tire's maximum is at 1 G, you can pinpoint on the graph the maximum abilities of this tire at four points. As the driver, you can choose to use the tire at less than its maximum if you like. At point A, for example, you would be using a portion of the tire's braking potential. By pressing harder on the brake pedal, your level of force would travel up on the graph, closer to the tire's maximum limit. The same is true of cornering. You could choose to use a portion of the tire's cornering ability, represented by point B; or by driving faster or cornering on a tighter arc, you could use more of the tire's cornering ability, up to the maximum available.

You can, however, mix abilities. Tires can develop a combined accelerating force and cornering force. Still, they won't provide you with cornering force if you are using maximum acceleration. At point A on the graph in Fig. 5-6, you would have zero potential for cornering. If you accelerated at a point short of maximum, the tire would have some traction left over to provide some cornering potential.

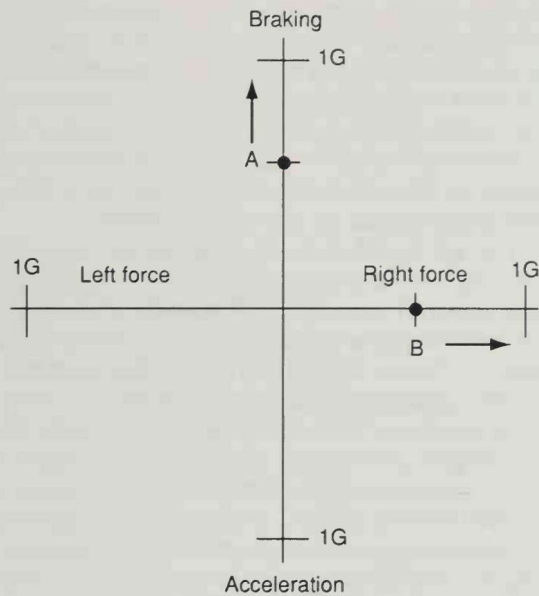


Fig. 5-5. A "friction circle" graph identifies the tire's maximum capabilities in the three forces it is capable of producing: accelerating grip, decelerating traction and cornering force.

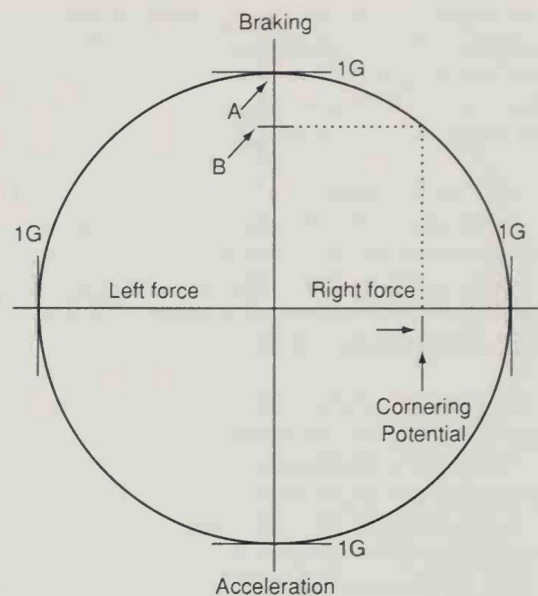


Fig. 5-7. By graphing all of the combinations of braking and cornering effort the tire can produce, you form a friction circle.

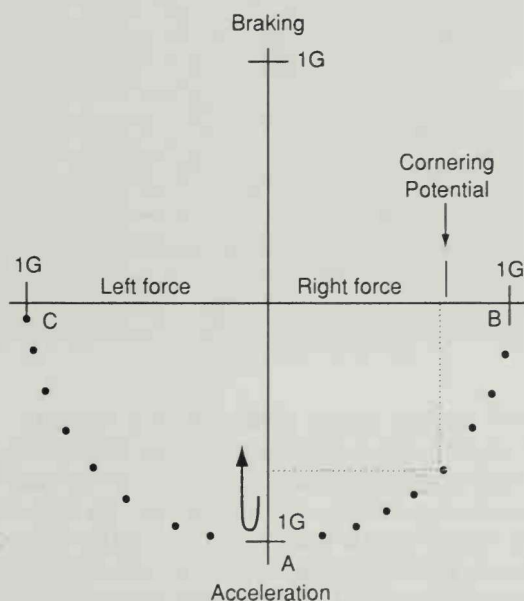


Fig. 5-6. By plotting all the combinations of Acceleration and Cornering force a tire is capable of, the lower half of the graph becomes a semi-circle.

If you graphed all of the combinations of accelerating and cornering force available, those points would form a semicircle, starting at point B, going through point A, and ending up at point C.

Using the Friction Circle

The friction circle allows you to think about what happens to one ability of the tire as you increase or decrease the demands for another ability. A lot of this is intuitive, but the graph helps you visualize what is happening with the tire. The bottom half of the circle in Fig. 5-6 represents the accelerate-turn combinations. The braking and turning forces available from this tire are represented in Fig. 5-7, as another semi-circle above the x-axis.

What you now have in Fig. 5-7 is the complete friction circle, a graph of all the potential combinations of ability available to you. For a moment, take a look at the brake-turn part.

Under threshold braking, at point A, there's no cornering force available. But if you relax the brake pedal effort so that your demands for braking force are reduced, say, to point B, you can trace a horizontal line out to the circle and see that, with this reduced demand on the tire's braking force, you get a considerable amount of cornering potential. The key, again, is not to identify specific points but to understand how you can affect the cornering potential of the car by the pressure on the brake pedal.

More Than Just a Concept

The friction circle is a real thing, not just some conceptualization in a text book. Our data collection system has the capability of drawing graphs which plot lateral G vs. longitudinal G—in essence, drawing friction circles for a real car on a real racetrack. Fig. 5-8 is a direct copy of one of these graphs drawn for the

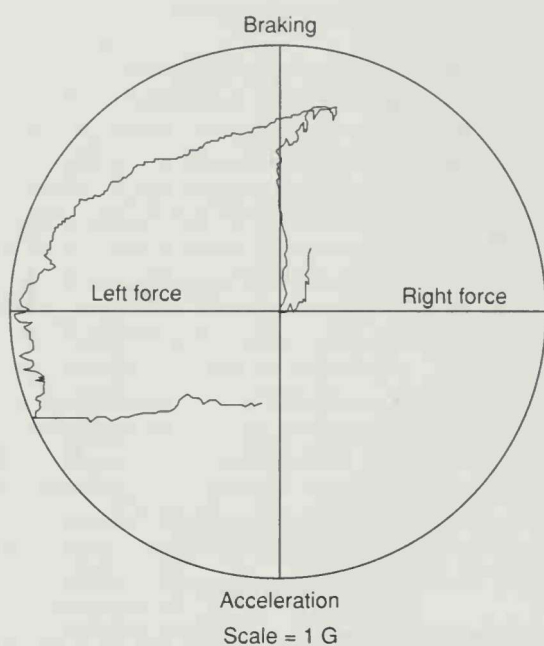


Fig. 5-8. An actual 1 G friction circle graph of a Formula Dodge pass up to and through the Sebring Hairpin.

braking entry of the Sebring hairpin. The traces are exactly as Mark Donohue theorized.

How Much Braking and Turning?

Corner entry technique as taught at many other racing schools would not include the topic of carrying braking past the turn-in point. We have found that in most racing situations, it is advantageous to do so. The question is not *if* you're going to do it, but *how*.

There is a wide range of brake-turning efforts. At one extreme there is the aggressive brake-turn right down to the apex, which you might use either at the end of a long straight leading to a compromise corner or in a passing maneuver. At the other extreme there is the light modulation off the brake pedal that you might use just past the turn-in for a fast sweeper.

Significance of the Throttle Application Point

When trying to choose the appropriate degree of brake-turning, an important variable is the location of the throttle application point. The later you apply the throttle, the more you'll use brake-turning.

As discussed earlier, you should begin the acceleration through the apex and onto the following straight as early as possible. This yields higher corner exit speed and consequently greater straightaway speed. The end result is improved segment times from the apex of the corner to the entry of the following turn.

If this is the case, why not begin accelerating *at* the turn-in point—you can't get any earlier than that! The

problem is that, depending on the acceleration potential of the car and the radius of the corner, starting the acceleration that early may create so much speed that you can't make the second half of the turn. As a rule, the throttle application point is where you can begin acceleration *and still make the corner*. If you have to feather the throttle at the exit of the corner to stay on, then you've begun acceleration too early.

You will find that, for the same corner, lower powered cars will have earlier throttle-application points than cars with more horsepower and greater acceleration potential. The speed range of the corner also affects the throttle application point since the car's ability to accelerate changes with speed.

In one of our Formula Dodges, for example, the car can gain over 5 m.p.h. per second in the 60 to 70 m.p.h. range, but this falls off to less than 2 m.p.h. per second between 90 and 100 m.p.h. The primary reason for this drop-off is an increase in aerodynamic drag as speed increases. As the car goes faster, more horsepower is used just to push the air out of the way, leaving less for acceleration. As a result, the throttle application point will move earlier as the speed of the corner rises.

"The amount you trail-brake into a corner really changes with the type of car you're driving. With a low powered car with a lot of grip, like a Formula Vee on slicks, there's not a lot of braking and turning because in most corners you're on the power so early in the corner. In the WSC car, for example, you use it a lot because with so much horsepower you can't get to the power early so you decelerate deeper into the corner."

—Jeremy Dale

How Corner Angle Affects Brake-Turning

The total amount of direction change required in a corner affects the throttle application point and therefore the extent of the brake-turn. Take a look at the three 55 m.p.h. corners in Fig. 5-9. The radius of the arc required to negotiate each corner is in the 200 foot range and a 1 G car can reach 55 m.p.h. on this arc. As usual, the radius would expand at the exit, as would the speed, but the constant radius arcs we've drawn will be fine for illustration purposes.

In each of these three cases, let's say that the *earliest* you can manage to begin acceleration away from the corner is 225 feet from the exit. Any earlier and the car falls off the road; any later and exit speed is lost.

In corner A, a 75-degree corner, you'll find that the 225 feet takes up 65 degrees of the change of direction. The car only needs to change direction 10 de-

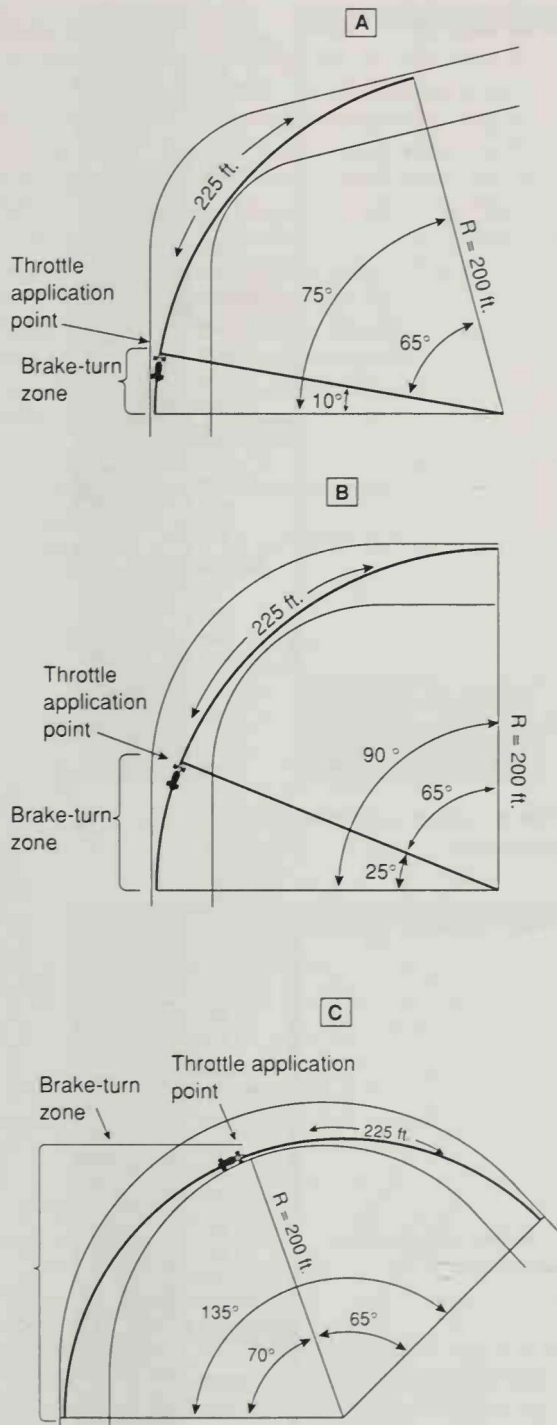


Fig. 5-9. The throttle application point varies with the total amount of direction change required in a corner.

grees before the throttle application point. The brake-turning portion of the corner entry is going to be brief if not nonexistent. Since the throttle application

point is barely 30 feet into the corner, you may choose to go directly from straight line deceleration (Block 2) to the brake-throttle transition (Block 4). In this case, in the 55 m.p.h. range it takes barely .4 seconds to go from the turn-in to the throttle application point, so you can skip the brake-turn phase.

In example B you have a 90-degree corner. The throttle application point is still 225 feet from the track out, but the difference is that the distance from the turn-in point to the throttle application point is now 86 feet. It would be wise to use some brake-turning to cover this distance at the highest possible average speed.

In corner C, as the corner angle increases to 135 degrees, the throttle application point moves further into the corner, right down around the apex. The length of the arc from the turn-in point to the throttle application point is over 240 feet. In the 55 m.p.h. range the car will be in the zone between turn-in and throttle application for around three seconds. The way you balance the car on a combination of braking and cornering will be very different than the braking technique used in the other two corners.

Pedal Force During the Brake-Turn

There are two basic ways of using the brake pedal in brake-turning. The first is to constantly relax the pedal pressure as the car approaches the throttle application point.

Let's say, for example, that at turn-in pedal pressure is at 140 lbs., and the brake-turn portion of the corner entry is going to take .7 seconds, as it does in this run into the carousel in Fig. 5-10. In a uniform brake modulation, you would release 20 lbs. of pressure each tenth of a second, going from 140 lbs. to 0 lbs. in a steady progression. When you take a detailed look at the brake pressure graphs of different drivers and different cars, you'll most commonly see this kind of brake-turning. In fact, the term "trail-braking" undoubtedly comes from explaining the process of trailing away the braking loads as the car decelerates while turning.

The second general way you can use the brakes while turning is by relaxing the brake pedal effort to a certain level and holding it there. In the example shown in Fig. 5-11, the driver turns in with 140 lbs. of pedal force and, over the first .3 seconds, relaxes the pressure to 70 lbs., then holds it there for a bit before relaxing the last 70 lbs. We have found that in corner entries where the brake-turn segment is longer (as in the 135-degree corner we looked at earlier), you try to find a specific combination of braking and cornering effort that slows the car *and* puts it on the right path for the apex. Once you've found the level of pressure that results in good speed loss and a solid cornering balance, you stick with it up to the point where the brake-throttle transition needs to take place:

Connected Corners

Another corner entry where you see this kind of brake-turning is where there is little or no opportunity to brake the car in a straight line approaching the corner. Since most or all of the slowing is done while turning, you try to brake at a level that accomplishes the speed loss but still leaves enough cornering force.

These situations are often found where there is a string of two or more corners, and the last corner is slower than the previous one. In Fig. 5-12, the left-hand corner leads to the right hander without an intervening straight. The curved braking zone will require braking and turning at a constant level of pressure.

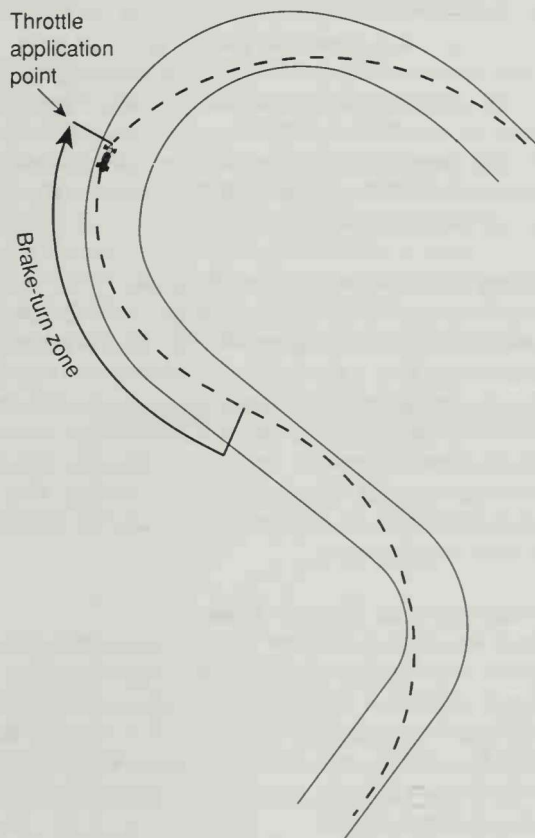


Fig. 5-12. Where there is no straight line braking segment, a constant level of brake-turning, subtly done, mixes deceleration and cornering.

This type of corner entry is tricky. At the beginning of the braking zone the car is already turning right. If you snap off the throttle on the way to the brake pedal, it is likely that the tail of the car will slide out toward the left. The throttle-brake transition needs to be done smoothly and over time to reduce this tendency.

The same goes for the initial brake application—smooth and progressive is the key, building to a con-

stant level until it's reduced in the brake-throttle transition. This slowing-down process is going on quite close to the edge of the racetrack, and if, for one of any number of reasons, you get in a little fast and the car wants to go straighter than you intend, you have to modulate the braking loads to create some cornering force to keep the car on the road. This is a standard part of braking and turning—being able to control the degree of cornering force available by increasing or decreasing the brake pedal pressure.

Ovals

Although we're concentrating primarily on road racing, drivers progressing up the ladder toward Indy Cars will encounter oval racetracks and their special demands. Entering long corners like those on ovals is very much like the situation in Fig. 5-12. The throttle application point, especially with more powerful cars, is relatively late in the corner, and there is significant lap time to be gained by carrying straightaway speed into the corner entry. The transition off the throttle onto the brake has to be done delicately, and many drivers smooth out this transition by learning to use their left foot on the brake pedal. For those of us accustomed to braking with our right foot, this is a skill that takes time to develop, but one worth learning if oval track racing is in your future.

"On ovals, most of the time, you drive the car down into the corner on power, then slow it down, then pick up the throttle for the exit. The car is already light down in the middle of the corner, so it's a delicate deal—I had to learn to use my left foot on the brake and, until you do it for a while, you find that you have a lot less feel with your left foot. It took a while to get good at it."

—Danny Sullivan

Shape of Corners

In decreasing-radius corners that *are* at the end of straights you are most likely to see constant-level braking and turning. This is because the more gentle arc in the first part of the corner often puts the turn-in point past the point where the road curves.

In Fig. 5-13, the car is first turned to match the arc of the edge of the road (point A), well before the turn-in point where the car is committed to the apex (point B). Since it is a long way between turn-in and the throttle application point, it is likely that you'll use constant-level braking and turning. Whether this will follow straight-line braking depends upon the total speed loss required. If the speed loss requirements are high, the process will likely start with threshold brak-

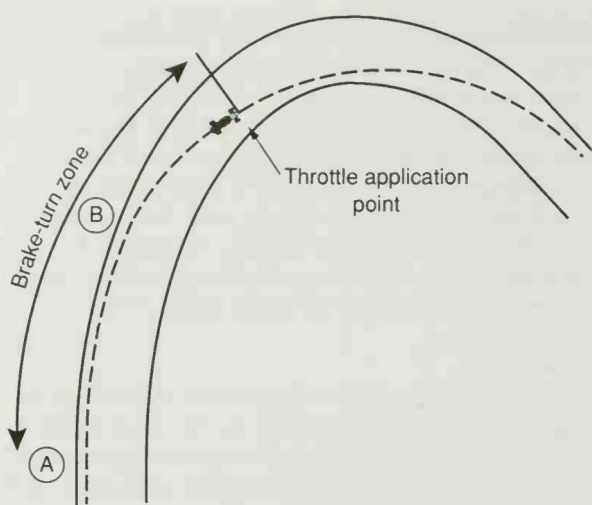


Fig. 5-13. Constant level brake-turning is more likely used in decreasing radius turns.

ing. Then, at the first turn of the steering wheel, reduce the brake pedal effort to a level that allows the car to negotiate the arc between A and B. At the real turn-in, the level may have to be reduced again to accommodate the tighter arc of the line toward the apex. In both segments, though, the pressure is likely to be constant-level rather than a bleed-off.

As speed loss requirements decrease, the braking point gets closer and closer to the corner. Especially in decreasing radius corners, the initial braking point may end up *in* the corner. If so, the cautions and requirements of starting the deceleration process while turning are the same as we outlined in the "connected corner" situation.

When Not To Use Brake-Turning

Starting to slow for a corner *after* you have turned into it is not the most common corner entry procedure, but it *is* one you should have in your repertoire. But there are times, even if you can do it well, when you might choose not to. The cases I'm thinking about are those which we dealt with when looking at small corner entry speed losses.

"I don't think I ever, ever, release the brake before the point of turn-in, but sometimes it's not so much to save time in braking. On fast sweepers, you don't want rotation, you don't want big weight transfer, you just want stability. So you knock off the speed before you get to the corner and have it as stable a platform as possible at the entry. It's a balance deal."

—Dorsey Schroeder

For example, if you only need to lose two m.p.h. of speed, why do it with a breathe on the straight when you could drive into the corner, brake-turn lightly, then transition back to power? The reason not to do this has more to do with handling considerations than with chasing the latest possible braking point. The sticky point is the throttle-brake transition.

When speed loss requirements are low, there isn't much time available to do a smooth throttle lift after the turn-in and, smooth or not, the initial lift off the throttle is going to tend to create oversteer. If the brake-turning process is going to be a long one, the application of the brakes settles down the oversteer by substituting a proportionately balanced loss of cornering traction, front to rear, for the unbalanced situation of using engine braking at the rear tires while the fronts are at full potential cornering traction.

For now, let's agree that you are going to carry some deceleration past the turn-in point of all corners where speed loss is required. In fast corners and small speed losses, this may mean the continuation of a throttle breathe ten feet past the turn-in. In some cases you will be losing 60 or 70 m.p.h. between the turn-in and throttle application point.

Using Throttle Before the Turn-In

As you've seen in the chapter on car control, simultaneous inputs of steering and throttle most often create understeer. This is the fundamental disadvantage of a throttle-then-steer entry. Once the understeer is created how is it defeated? Most often, a lift off the throttle, which comes *after* what would have been the throttle application point, takes care of this problem. The down side is lower exit speed caused by the delayed throttle application.

Brake-Turn vs. Corner Type

Your decisions about how much brake-turning to do are also affected by whether you're dealing with a Type I, Type II, or Type III corner. As you no doubt recall from the chapter on the racing line, we said that the line could change depending upon whether a corner led onto a straight, came at the end of a straight, or led to another corner.

In corners leading onto a straight, exit speed is king. You want to bias the throttle application toward the early side, so you would tend to do a shorter brake-turn in Type I corners.

In Type II corners where exit speed is not the prime consideration, you could make use of the braking and turning ability of the car and decelerate up to or past the apex, making up time in the braking zone.

In Type III corners, you are more likely to not only extend the brake-turning zone but to find yourself in situations where the slowing down process will have to begin while the car is still turning.

"Late in my career I drove a Porsche Carrera RSR at Watkins Glen and it was a beautifully balanced car—you could do anything with it.... My habit on the approach to the 90 at the Glen was to pitch the car at the entry, get a lot of opposite lock and try to be really aggressive with the power trying to get the best shot up the back straight. On one lap—I think I was passing someone—I really carried the brakes down toward the apex. The car was much more stable than usual and I found since it was more stable that I could apply more throttle sooner and the speed at the trackout was much better."

Car Capability Affects Brake-Turn

The handling predispositions of your car affects how aggressively you choose to brake and turn. If the car's setup is biased toward the oversteer side, whether intentional or not, the brake-turning aspect of corner entry has to be de-emphasized. This oversteer tendency may be generated in any of number of ways. It could be that the rear sway bar is too stiff, the rear ride height too high, the front sway bar broken or missing—any of a number of chassis-related thrillers. It

could be the rear tires going off or the brake bias proportioned too much to the rear. If the car can't be changed back toward neutral handling, you have to adapt. One of the adaptations for driving an oversteering car is to reduce the level of aggressive braking and turning, if not abandon it altogether.

In these situations you're trying to reduce the braking traction demands on the rear tires so that they can supply more cornering force, settling down the oversteer. You'll find that, likewise, the accelerating and turning ability of the car is going to suffer and that more sensitive throttle application will be necessary to keep the rear end behind the front.

BLOCK 4: THE BRAKE-TO-THROTTLE TRANSITION

The “brake-to-throttle transition,” a fancy way of describing how you take your foot off the brake and put it back on the throttle. Let’s look at the variables.

Time

The first variable is how long the process takes. You can take your own sweet time going from your initial braking level (be it threshold or not) down to zero brake pressure. You could also pop your foot off the brake pedal in a nanosecond.

Sometimes, as you can see in Fig. 5-14, you gradu-

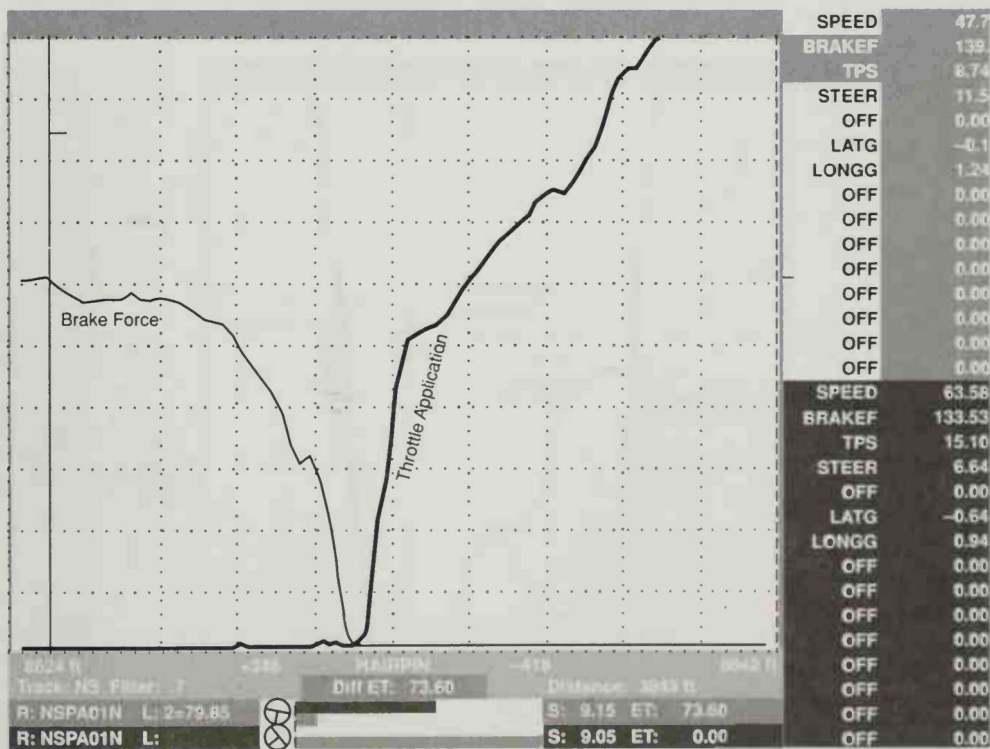


Fig. 5-14. In this graph of braking and throttle application at the Sebring hairpin, the driver gradually reduces braking force from 140 lbs. to zero over 1.1 sec. then makes an instantaneous switch from the last bit of braking to the application of throttle.

ally reduce braking effort down to zero and the right foot instantly picks up the throttle and begins to squeeze it on. This is exactly the same motion as a brake-turn done in the bleed-off style—that is, with a progressive reduction in brake pressure. It's the same as a slow, deliberate brake-throttle transition. The two are interchangeable.

If the brake-process is one where brake pressure is more or less constant, the brake-throttle transition can be slow and gradual as it is in Fig. 5-15, where it takes 1.25 seconds to go from roughly 80% braking down to zero. You *could* do it suddenly, as in Fig. 5-16, where everything else is similar but the brake-throttle transition is more abrupt.

Effects On Handling

The length of the brake-throttle transition will affect how the car sets its brake-cornering slip angle. A slow brake-throttle transition allows the reduction in braking effort to keep in sync with the changes in front-to-rear loading. These changing forces work together over time, maintaining (or at least not upsetting) the overall cornering balance of the car. A sudden, sharp reduction in the braking effort delivers instant cornering traction to the front of the car by suddenly giving *all* of the front tires' grip over to cornering force. The car is likely to point aggressively toward the apex, increasing the car's yaw angle. Sometimes you want this to happen, sometimes you don't.

In a sharp, slow corner, having the car rotate an extra 5 to 10 degrees might be helpful—it keeps the car out of understeer. In a 95 m.p.h. sweeper, however, you might not enjoy the kind of excitement that 10 degrees of extra yaw angle provides.

The Pause

The second element you can use toward the same end is to choose to go directly to the throttle to stabilize the yaw you have created, or you can pause before picking up the throttle.

The "pause" is an intentional trailing-throttle oversteer maneuver. If the car has some rotation going, but not enough for your liking, a pause before going to throttle will help the car increase its yaw angle. If you like the extent of the slide you've got going, go immediately to enough throttle to stabilize the yaw angle. Once at this point, you're now playing the corner exit game of squeezing on as much power as you dare, trying to maximize your speed coming onto the straight.

Assumptions We've Made

The type of car you're racing will affect your braking choices. Many racecars, especially the ones that a beginning racer is likely to start with, have excellent brakes. On a purpose-built racecar like a Formula-Ford, Spec Racer, Ford 2000, Formula Atlantic, or Sports 2000 car, the brakes should be able to work at their maximum level every time you go to them, from

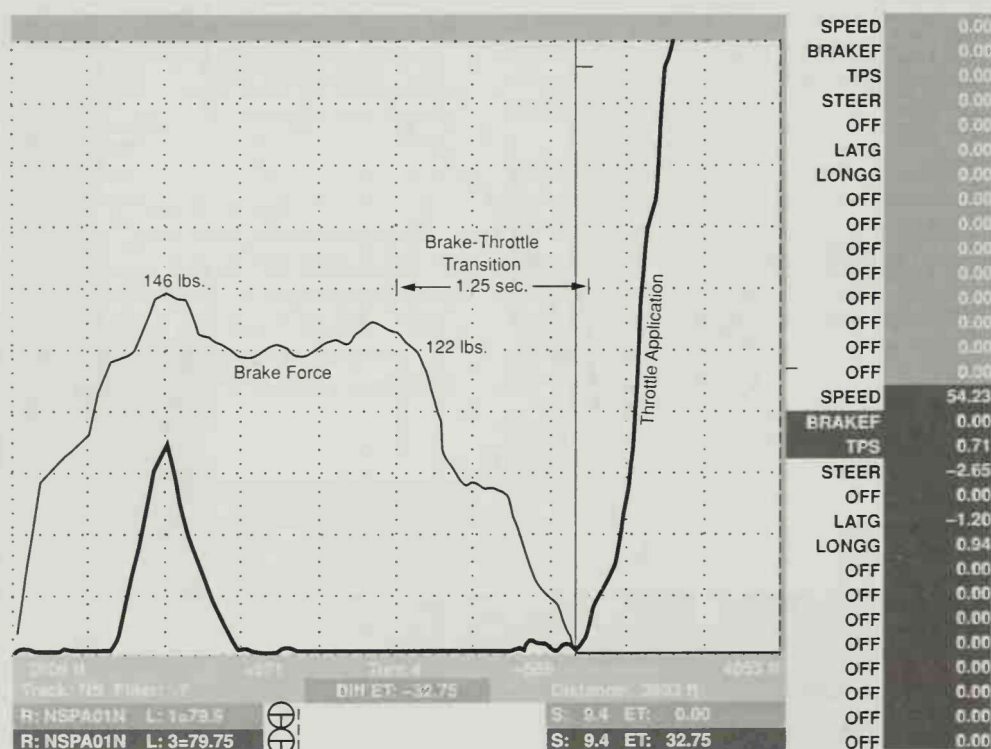


Fig. 5-15. The smooth brake-throttle transition in this example takes 1.25 seconds.

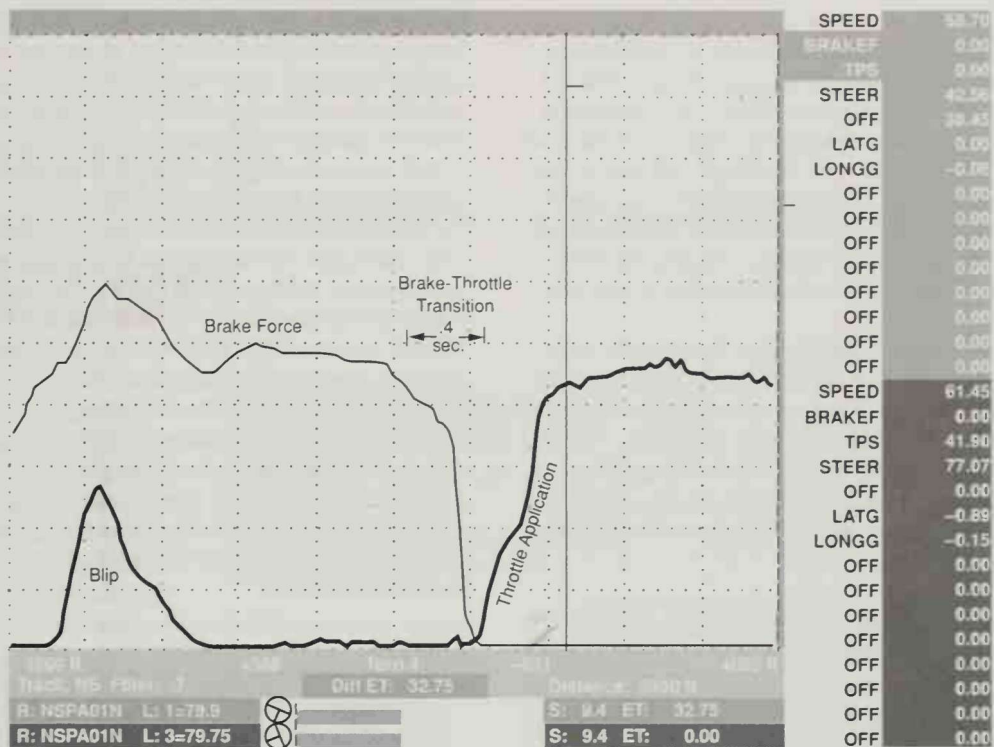


Fig. 5-16. Similar to Fig. 5-15, but the transition from brake to throttle happens in .4 seconds, which affects car rotation at the corner entry.

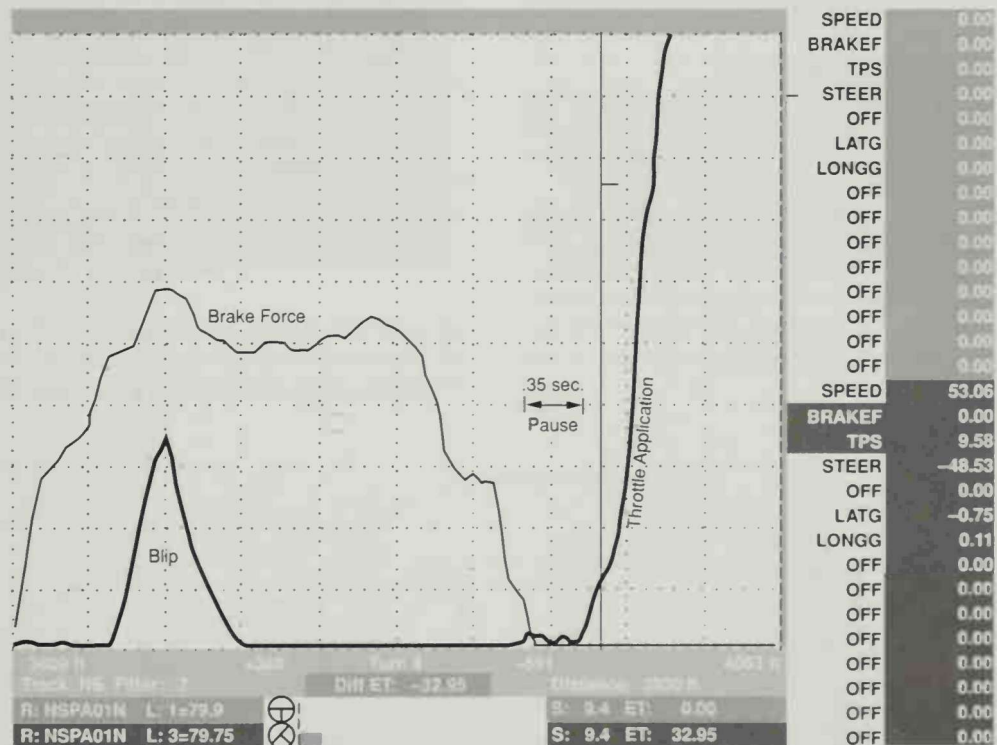


Fig. 5-17. Here the driver apparently didn't like the amount of yaw and paused for .35 seconds between the brake release and the application of plenty of power. Note the position of the lower steering wheel on the graph—it's turned significantly left in this right hand corner. The "pause" worked.

beginning to end of your race.

Expecting a showroom stock car to perform like a Formula car under braking is a big mistake. In a car designed primarily for around-town use, the brakes are not up to the task of stopping a heavy vehicle time after time for hours on end. Although threshold braking may be desirable and achievable for a few qualifying laps, the faster and heavier the car, the more likely it is that you will have to modify your approach to braking to compensate for deterioration in the braking system.

Remember, during a long race, the pursuit is not for the single fastest lap but for the fastest *average* lap time over the race distance. An intelligent racer has to make the distinction between the technique which is usable in a sprint race in a bulletproof car and the driving required in order to save a car which has deteriorating performance in one or more of its systems.

"Sure, you'd like to try and go into every brake zone as hard as you can, but in a long race you can't drive every one at 100%—98% is more like it. If something changes, and it always will, or if you run into problems, you have a one or two percent margin. It doesn't have to cost you speed."

—Jeremy Dale

The Key Points of Corner Entry

Corner entries are exciting—more exiting than corner exits, to be sure. Coming out of a corner, you feel more fully in control, squeezing on the accelerator. If anything unusual or threatening takes place, you just don't push the throttle down as aggressively. Coming into the corner you feel a bit more like a passenger.

You've made a commitment at your braking point that your technique this lap has to be very close to the last lap's or you'll arrive at the corner going too fast to make it. Consistency in brake point and the level of pressure used on the brake pedal is critical.

It's important to keep in mind what you're trying to accomplish at the corner entry. Remember that taking the speed off in the shortest amount of time is only part of the goal. Slowing down to the right speed—the maximum speed the car can support on an arc that makes the apex—is even more important because, especially in entry level cars with limited horsepower, this speed frequently determines the cornering speed and your speed down the next straight.

Different corners present different speed loss situations and you can count on the fact that you won't use the brakes the same way in every corner. A little thinking and planning before you go out in the race-car can help you prepare for improving your corner entry performance.

"A lot of people jump on the brakes very hard. I was always a guy who braked, for the most part, very easy. I didn't use up the brakes at all since I tended to roll off the throttle and onto the brake more easily."

"To put it in perspective, at Laguna Seca, which is hard on brakes, Rick Mears and I were teammates at Penske and Rick finished the race with only 70 thousandths of an inch of brake pad material left. I only used 70 thousandths of the pad in winning the race. People brake differently but can still run the same lap time, especially in a race."

—Danny Sullivan